

Motor Modeling

디지털제어 및 신호처리 - 2조

유민결 유채은 김솔

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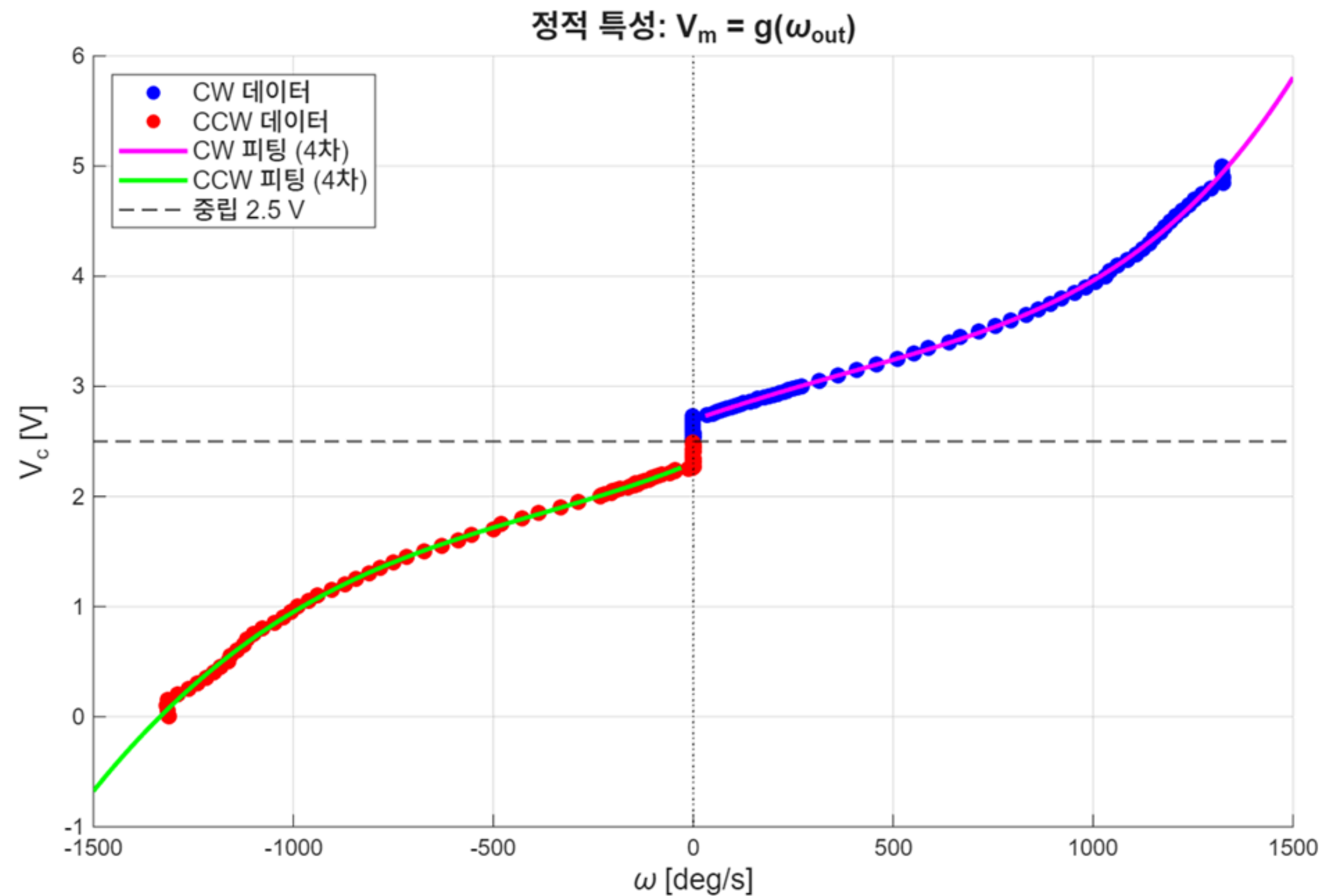
02. Dynamic Motor Modeling

03. Validation

1. Static Motor Modeling

Deadzone 및 Saturation 확인

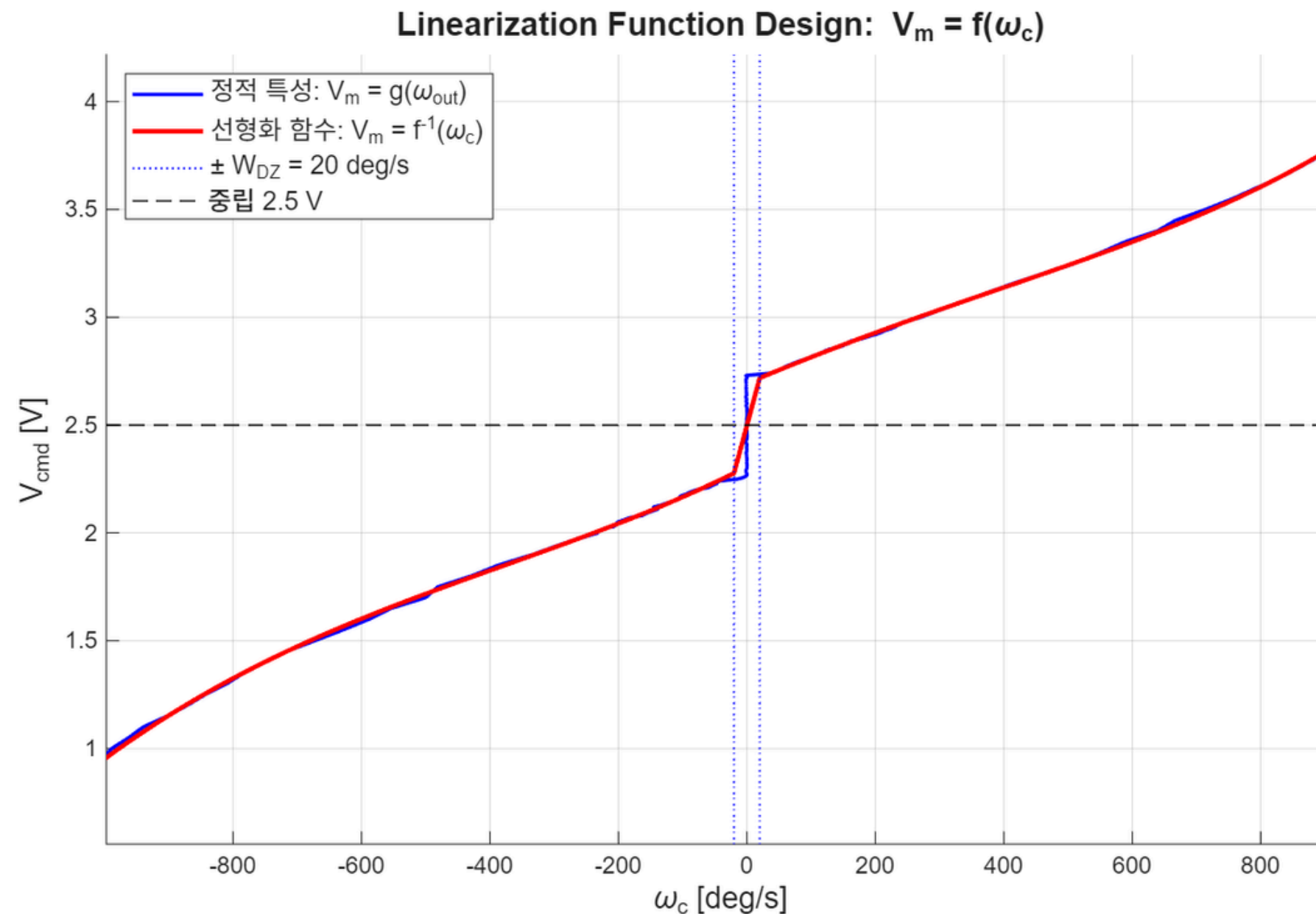
- $2 \leq V_c \leq 3$: 0.01V 간격
- $V_c < 2$ or $V_c > 3$: 0.05V 간격으로 data 수집



1. Static Motor Modeling

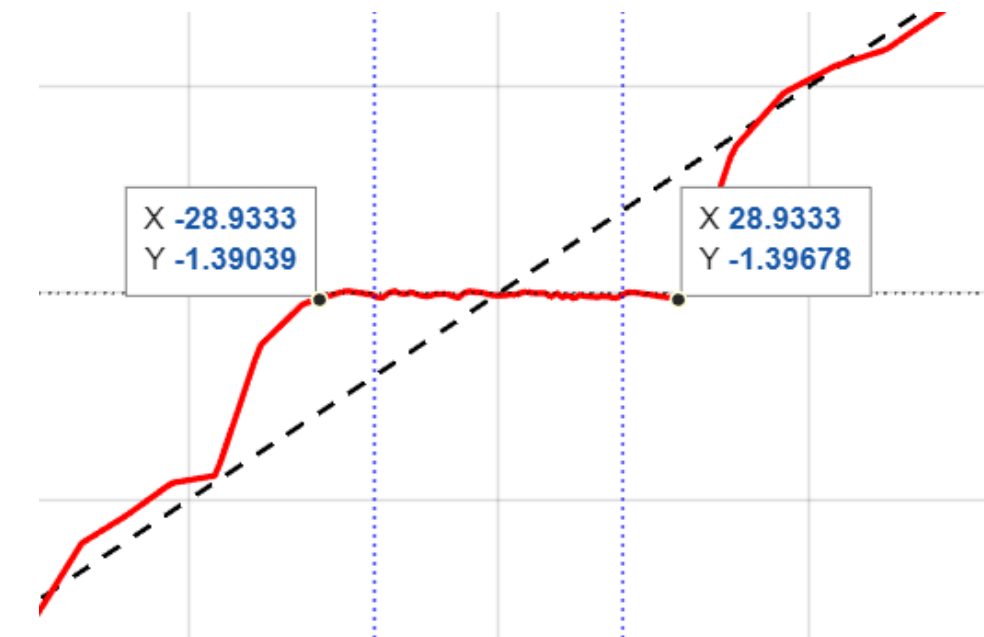
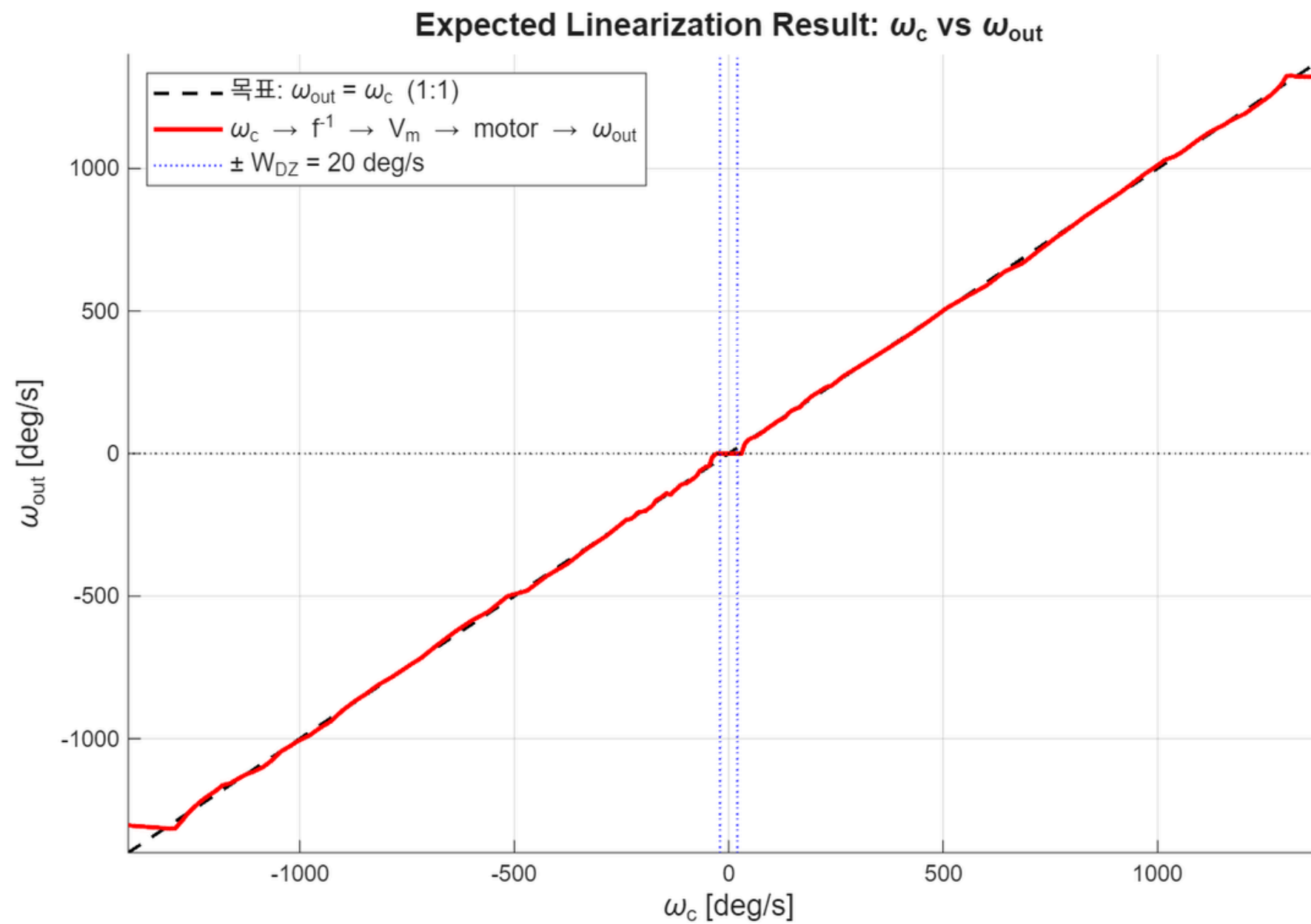
Linearization

- 선형화 범위(CW) : $30[\text{deg/s}] \leq \Omega_{\text{cmd}} \leq 1200[\text{deg/s}]$
- 선형화 범위(CCW) : $-1200[\text{deg/s}] \leq \Omega_{\text{cmd}} \leq -30[\text{deg/s}]$
- CW, CCW 각각 4차로 fitting
- 역함수에서 데드존 구간은 직선으로 연결



1. Static Motor Modeling

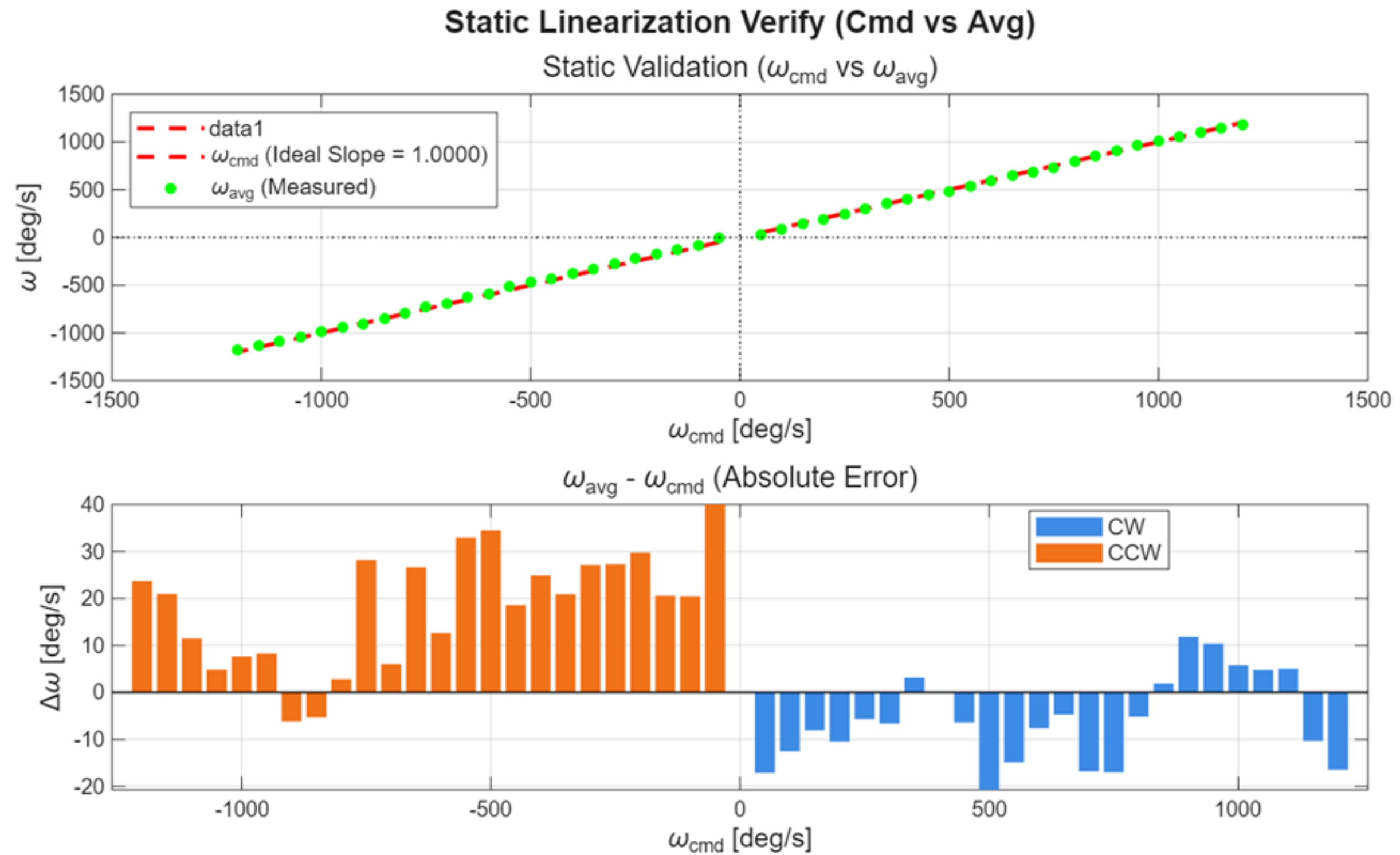
Linearization Result



- OmegaCmd_DZ
- (CW) : -28.93[deg/s], (CWW) : 28.93[deg/s]

1. Static Motor Modeling

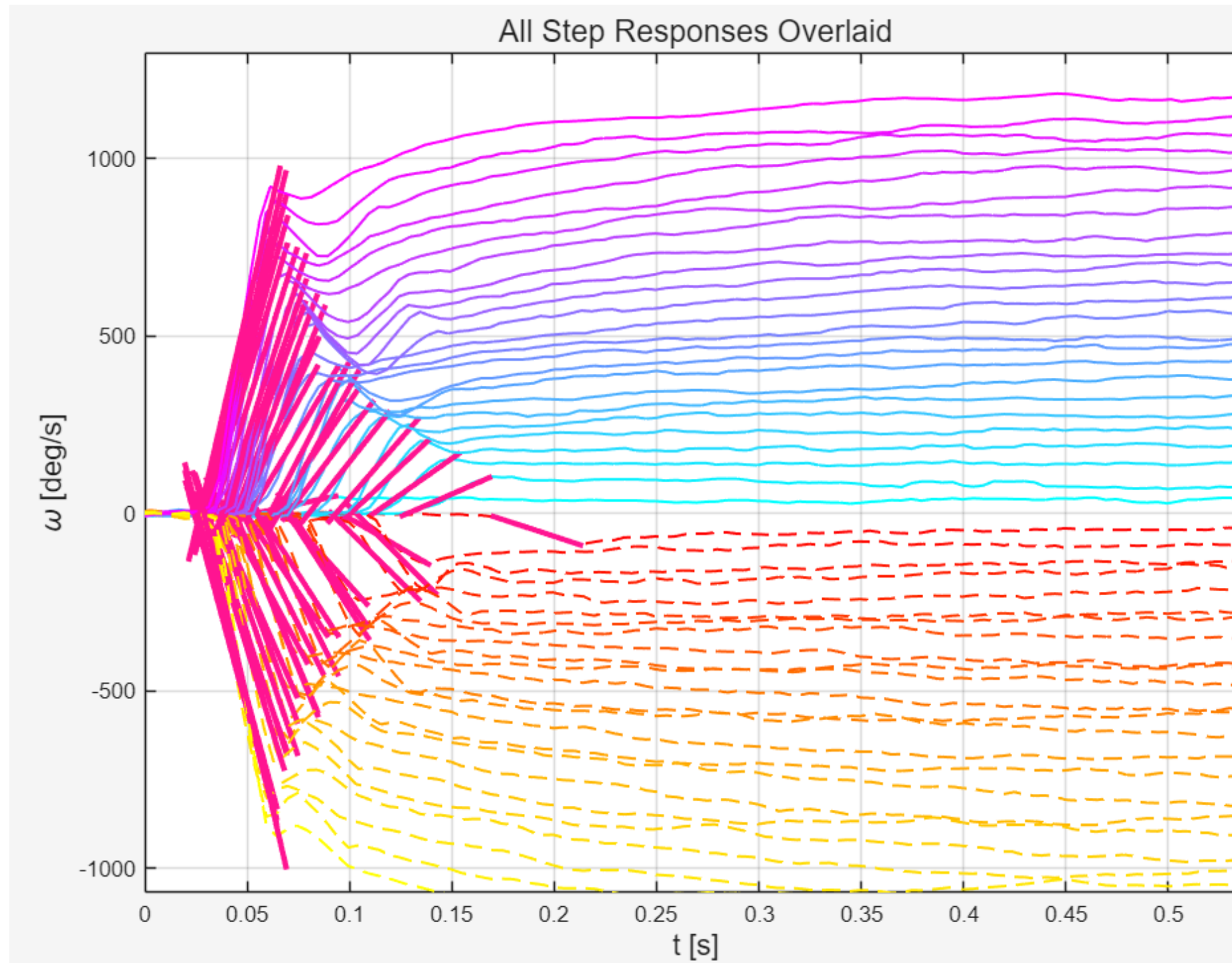
Static Validation



1. Static Motor Modeling

Slewrate Limit 확인

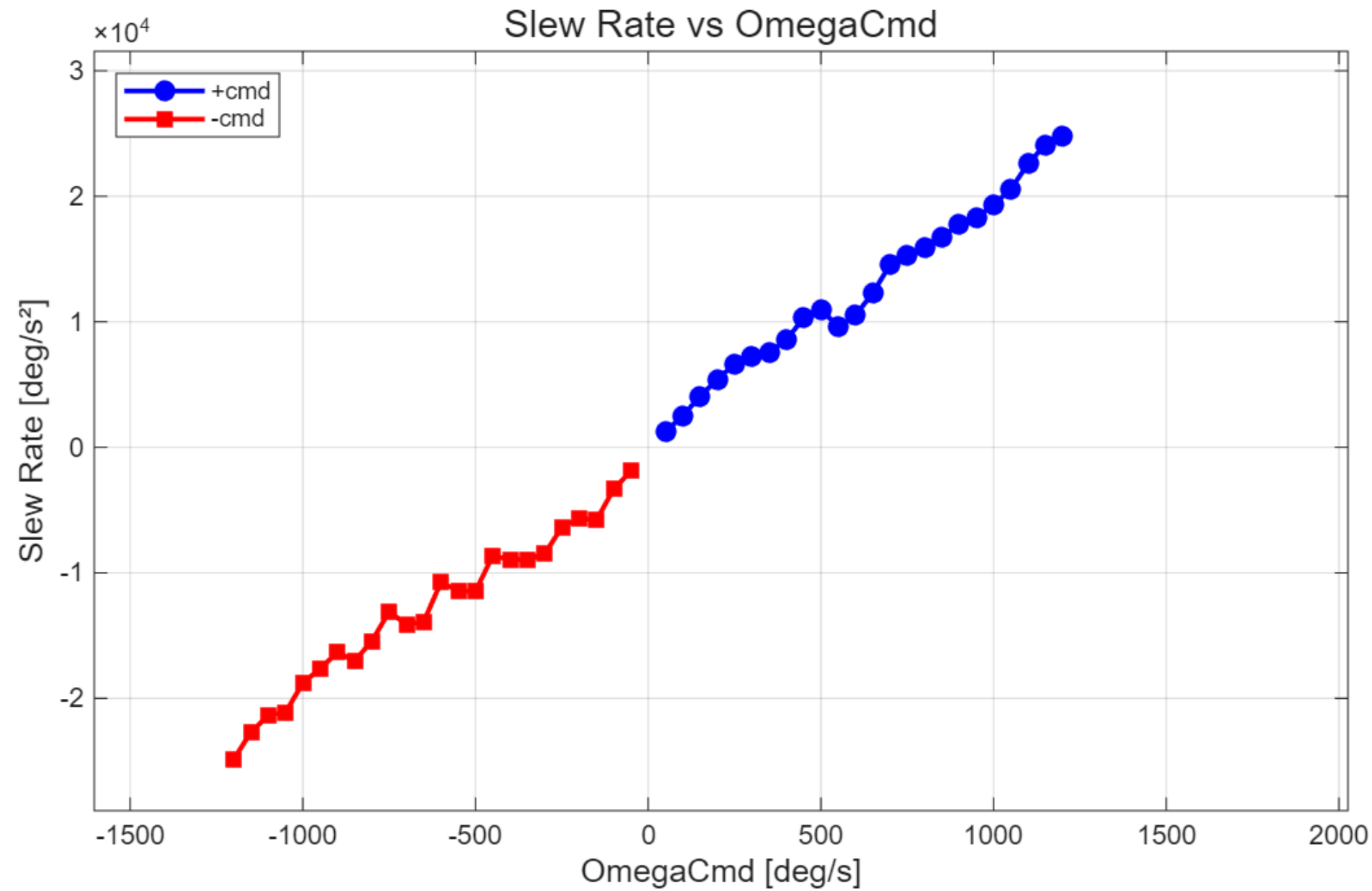
- 측정 범위: $\Omega_{Cmd} = \pm 50 \sim \pm 1200 \text{ deg/s}$
- 각 스텝 응답의 초기 과도구간에서 최대 기울기 측정



+50 deg/s	Slew +350: +7570.6 deg/s ²
Slew +50: +1227.5 deg/s ²	-350 deg/s
-50 deg/s	Slew -350: -8924.5 deg/s ²
Slew -50: -1886.8 deg/s ²	+400 deg/s
+100 deg/s	Slew +400: +8558.2 deg/s ²
Slew +100: +2524.7 deg/s ²	-400 deg/s
-100 deg/s	Slew -400: -8970.8 deg/s ²
Slew -100: -3341.1 deg/s ²	+450 deg/s
+150 deg/s	Slew +450: +10298.6 deg/s ²
Slew +150: +4040.5 deg/s ²	-450 deg/s
-150 deg/s	Slew -450: -8614.2 deg/s ²
Slew -150: -5799.9 deg/s ²	+500 deg/s
+200 deg/s	Slew +500: +11004.1 deg/s ²
Slew +200: +5426.5 deg/s ²	-500 deg/s
-200 deg/s	Slew -500: -11473.9 deg/s ²
Slew -200: -5671.2 deg/s ²	+550 deg/s
+250 deg/s	Slew +550: +9602.2 deg/s ²
Slew +250: +6624.2 deg/s ²	-550 deg/s
-250 deg/s	Slew -550: -11441.0 deg/s ²
Slew -250: -6358.9 deg/s ²	+600 deg/s
+300 deg/s	Slew +600: +10518.0 deg/s ²
Slew +300: +7287.3 deg/s ²	-600 deg/s
-300 deg/s	Slew -600: -10741.7 deg/s ²
Slew -300: -8502.7 deg/s ²	+650 deg/s
+350 deg/s	Slew +650: +12282.7 deg/s ²

1. Static Motor Modeling

Slewrate Limit 확인

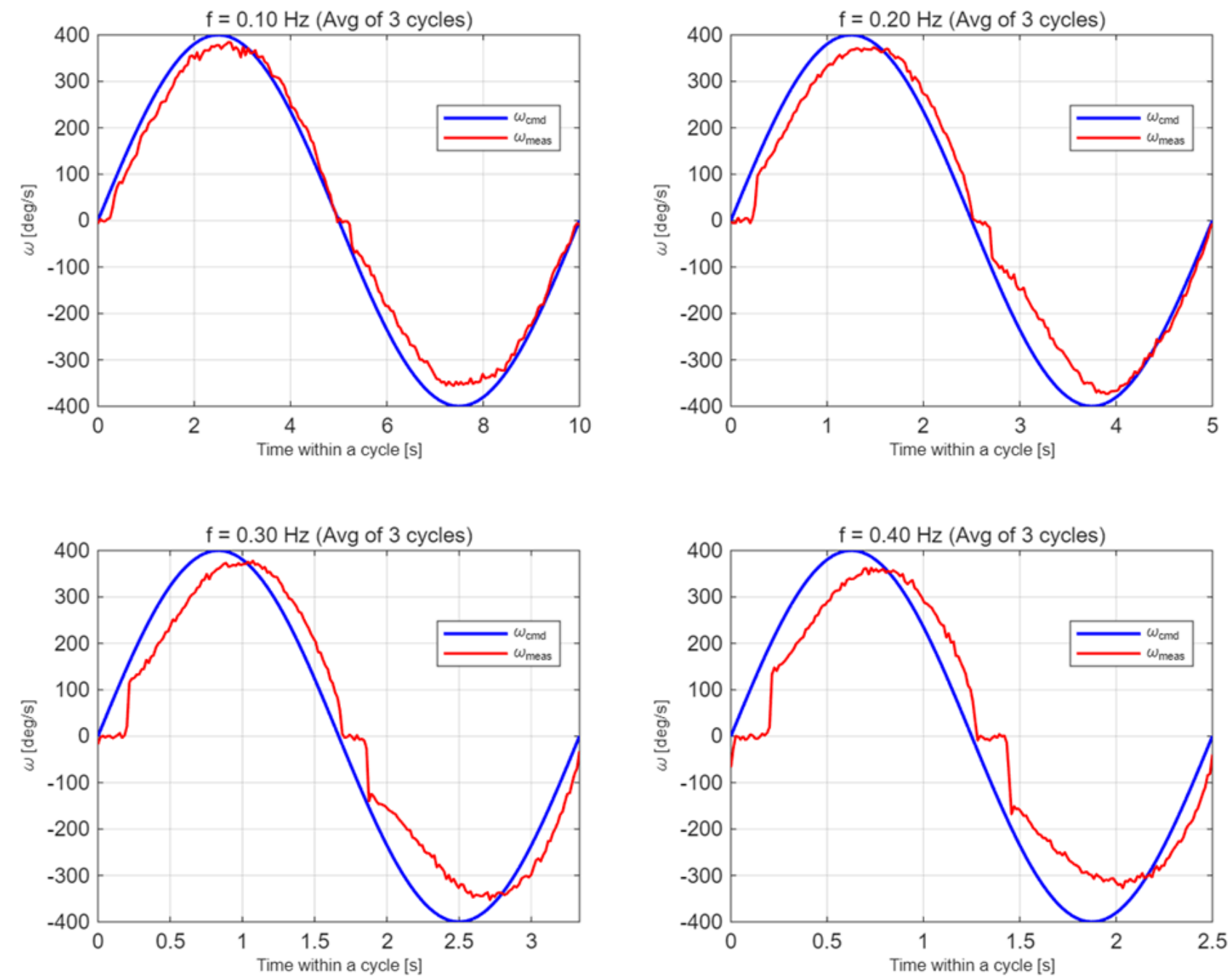


- 슬루레이트는 OmegaCmd에 비례하여 선형적으로 증가
- 특정 입력 크기 이상에서 포화되는 현상이 관찰되지 않음

2. Dynamic Motor Modeling

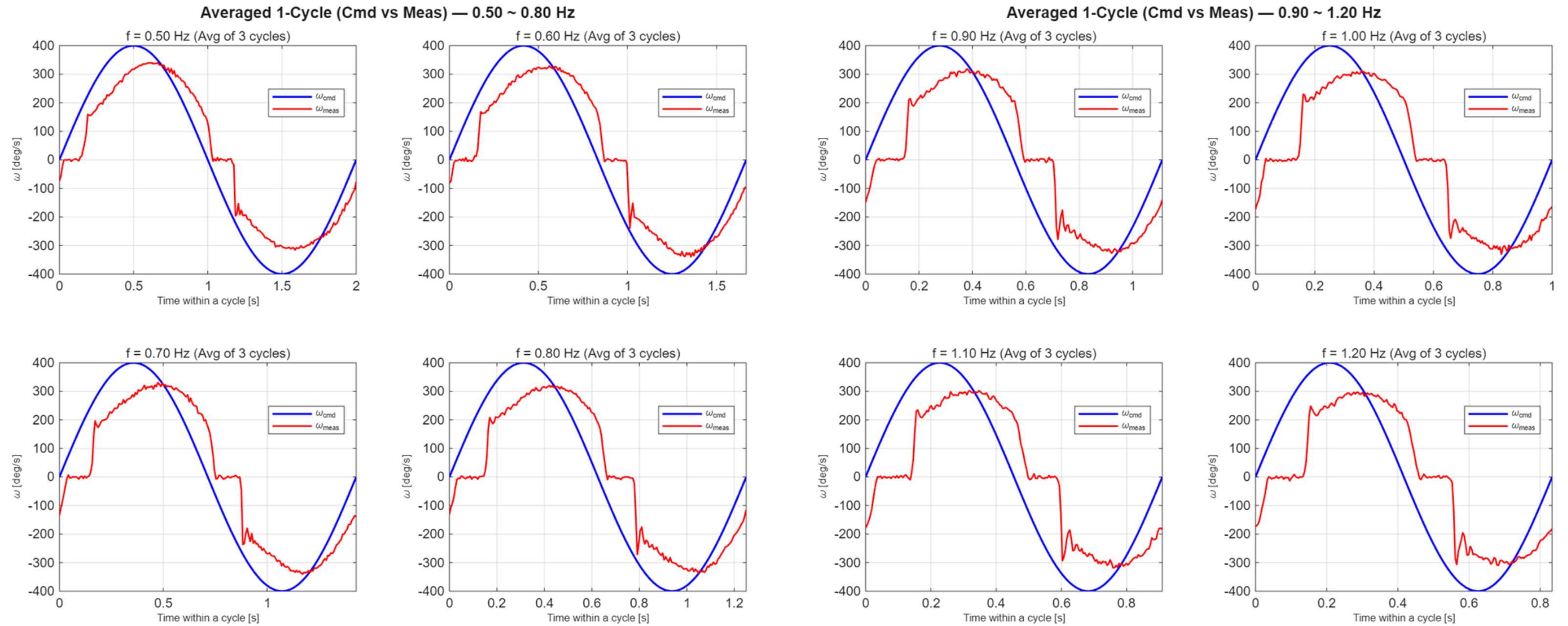
Frequency sweep

Averaged 1-Cycle (Cmd vs Meas) — 0.10 ~ 0.40 Hz



2. Dynamic Motor Modeling

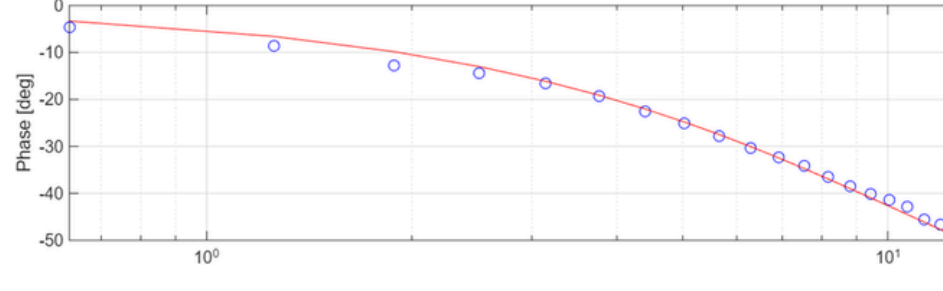
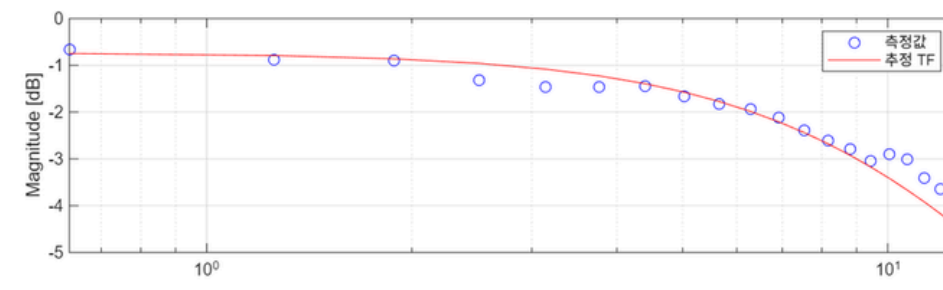
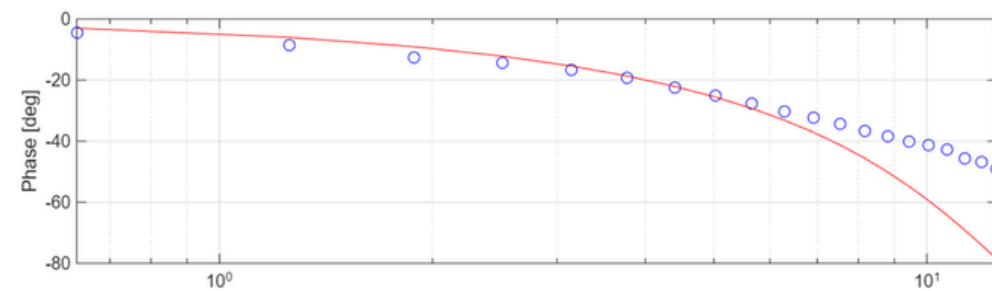
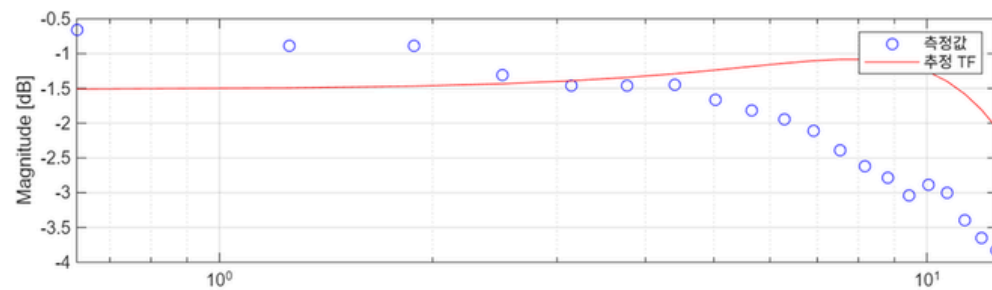
Frequency sweep



2. Dynamic Motor Modeling

Frequency sweep 및 모터 전달함수 추정

- Input freq: 0.1 ~ 2.0 [Hz]
- Amplitude: Omega_Cmd = 400deg/s

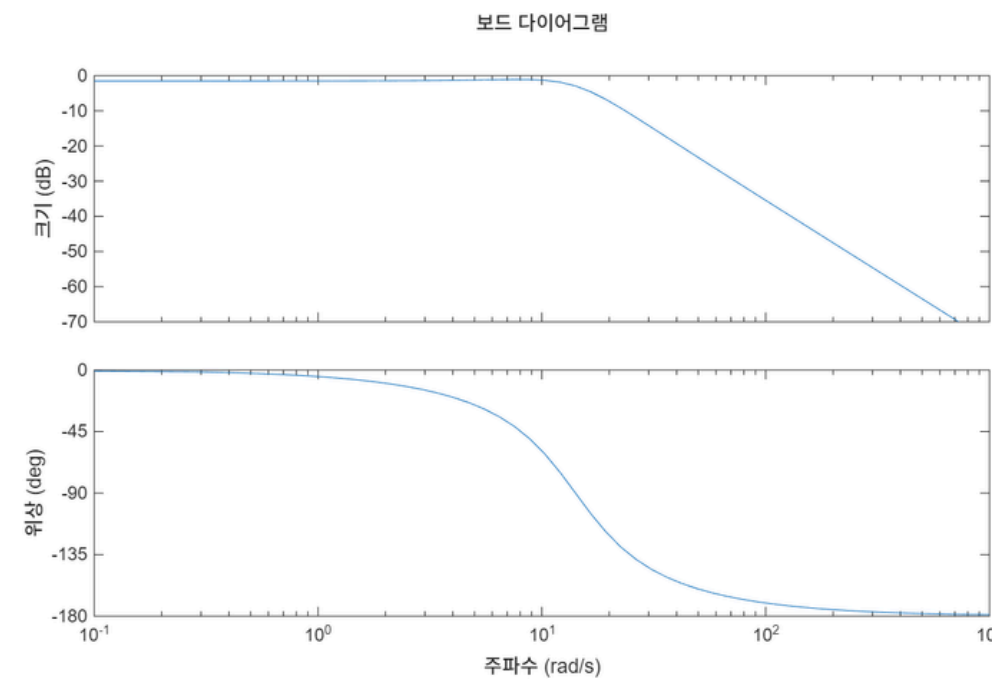


$$G_{m2}(s) = \frac{167.8}{s^2 + 16.62s + 199.7}$$

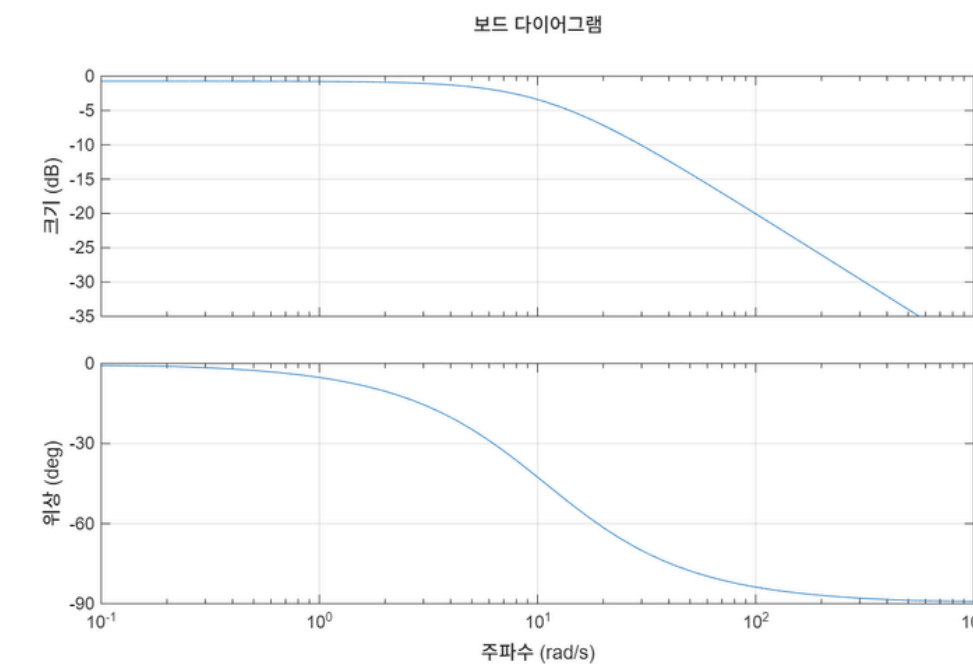
$$G_m(s) = \frac{9.993}{s + 10.87}$$

bandwidth = 1.71 [Hz]

2차



1차

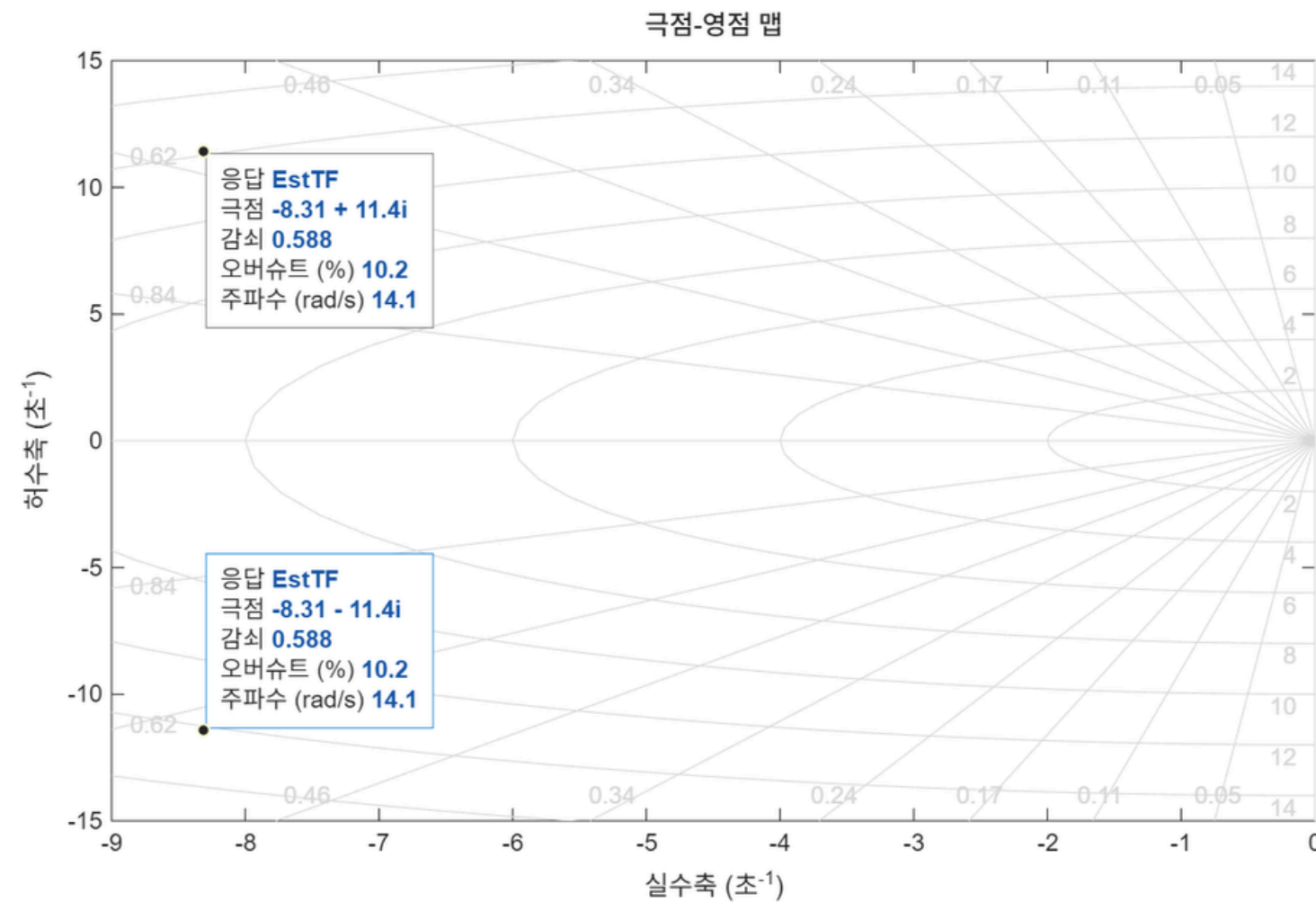


2. Dynamic Motor Modeling

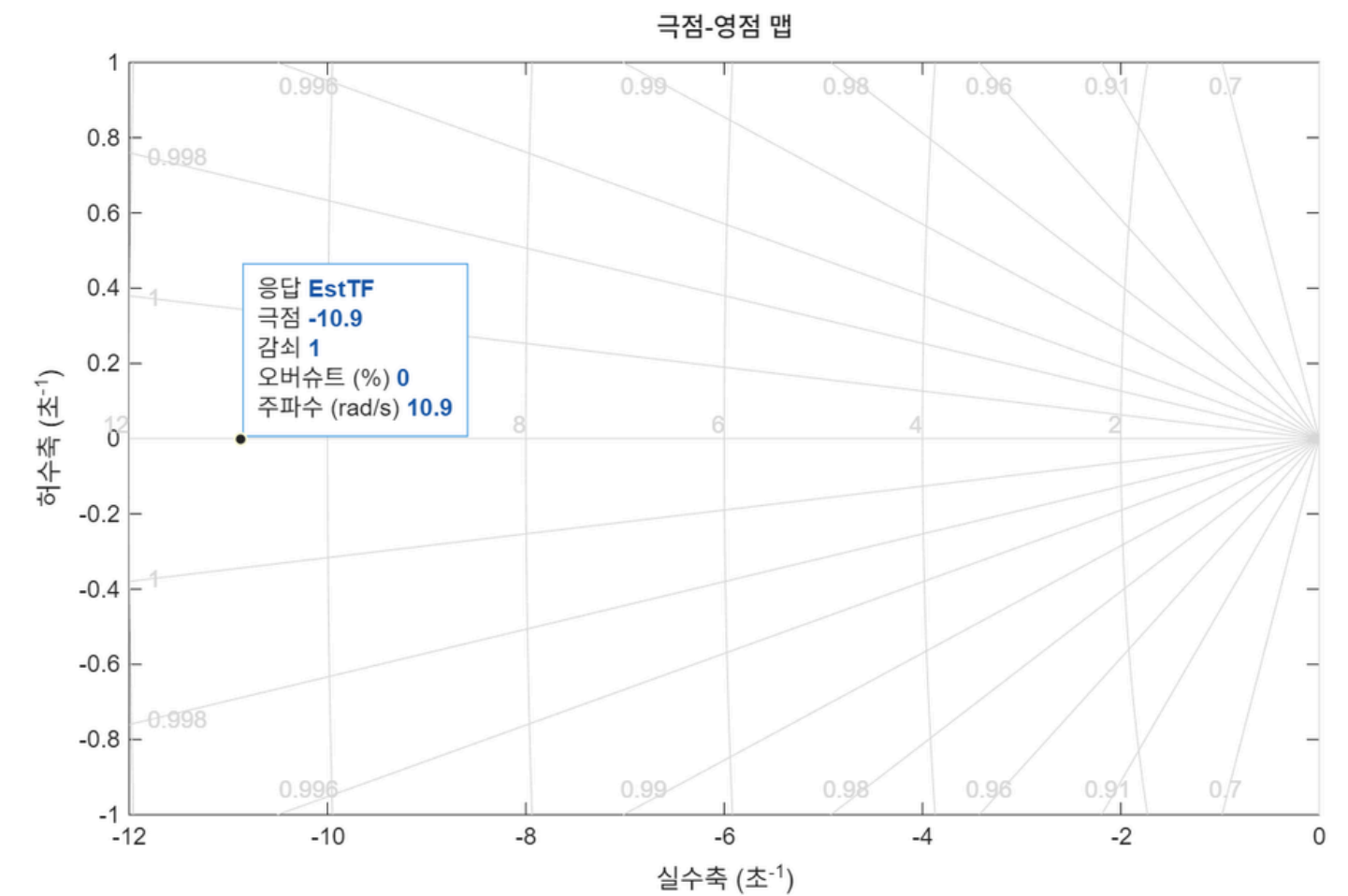
Pole-zero Map

$$G_m2(s) = \frac{167.8}{s^2 + 16.62s + 199.7}$$

$$G_m(s) = \frac{9.993}{s + 10.87}$$



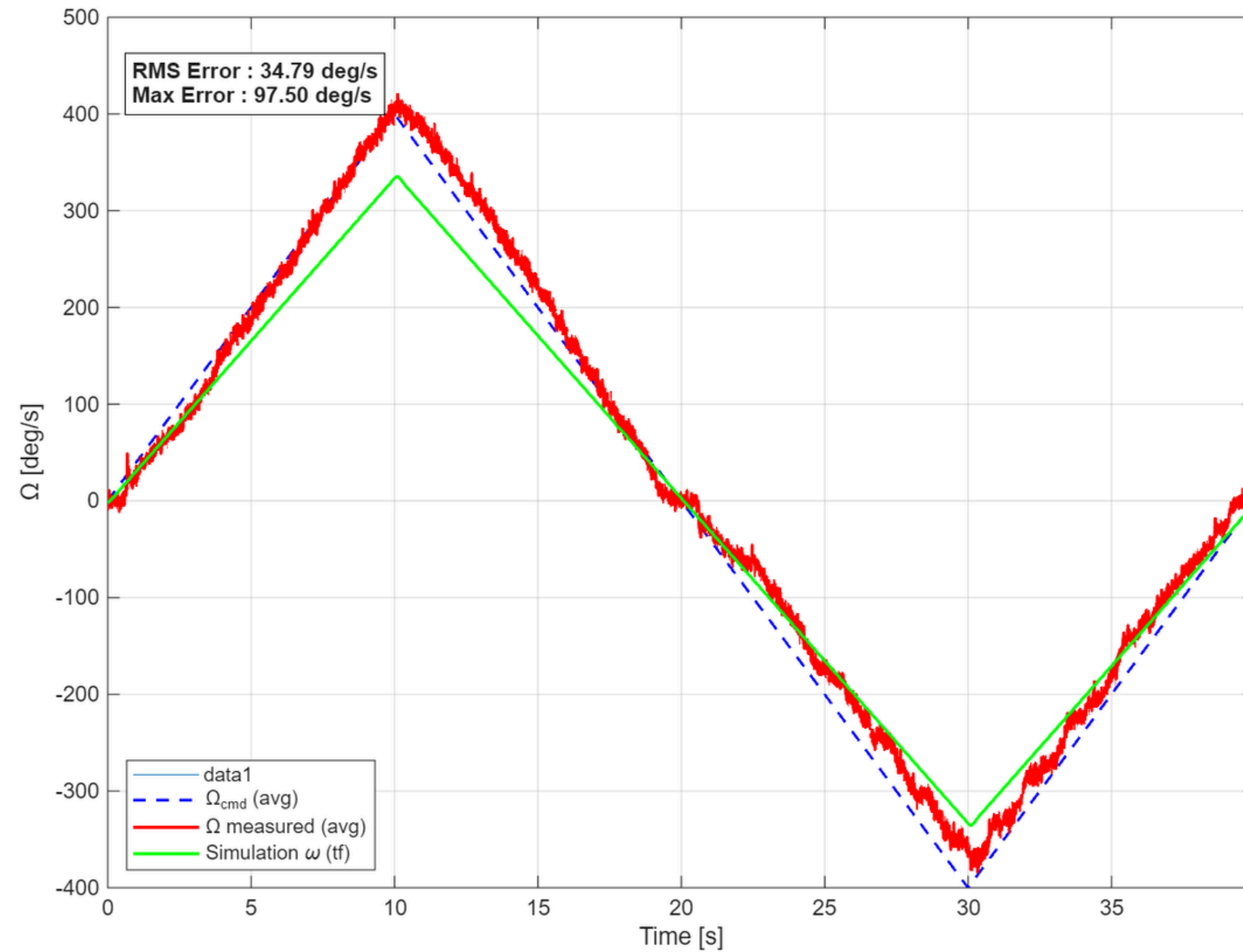
2차



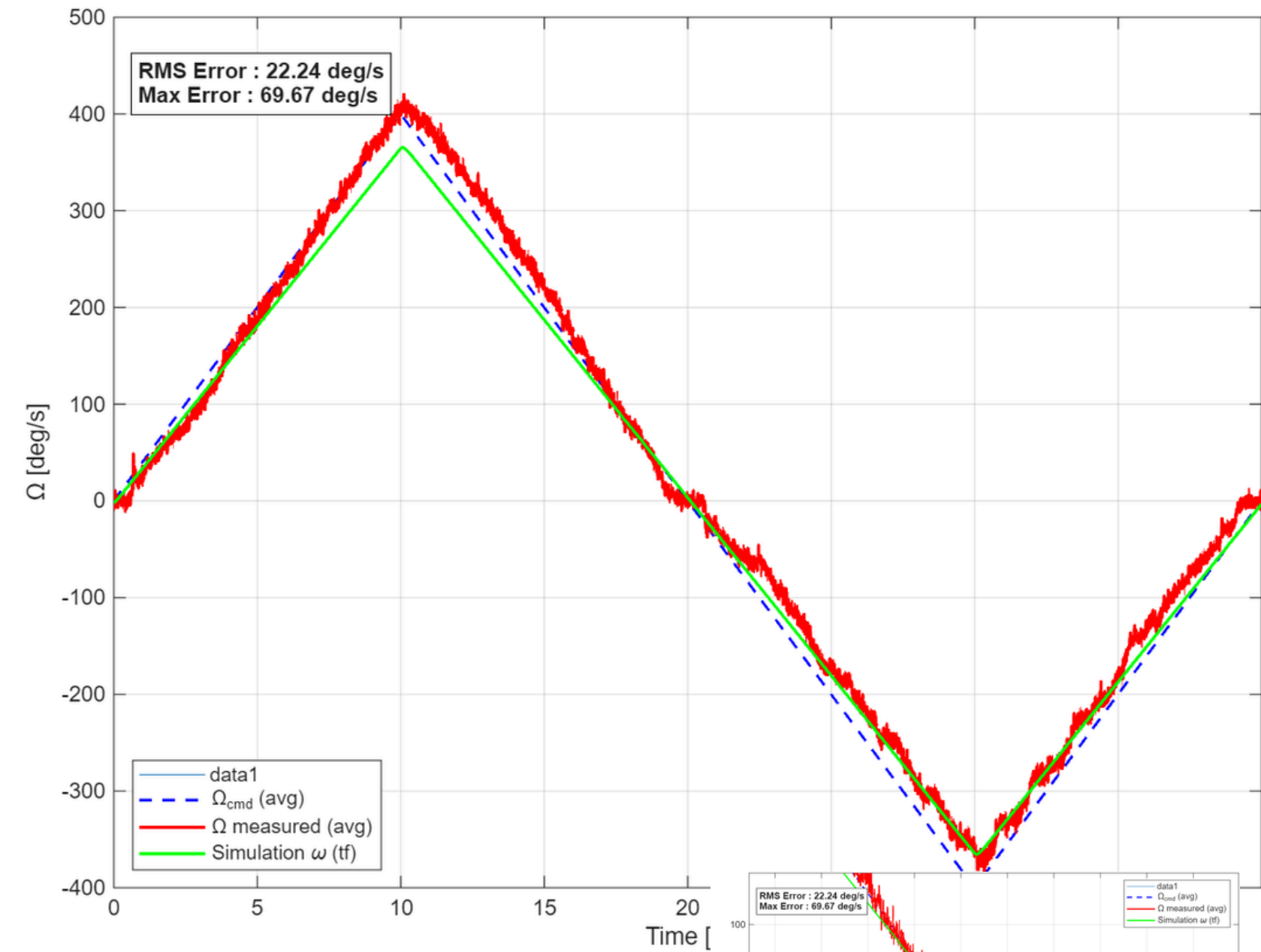
1차

3. Model Validation

- 삼각파 validation

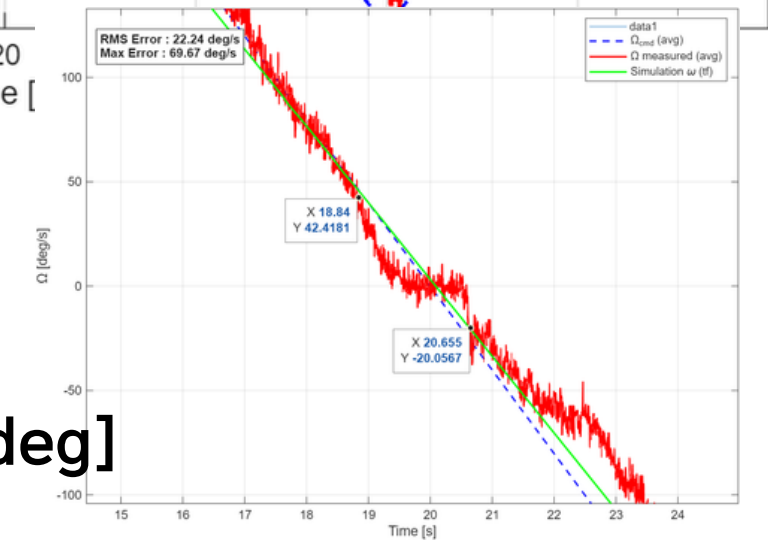


2차



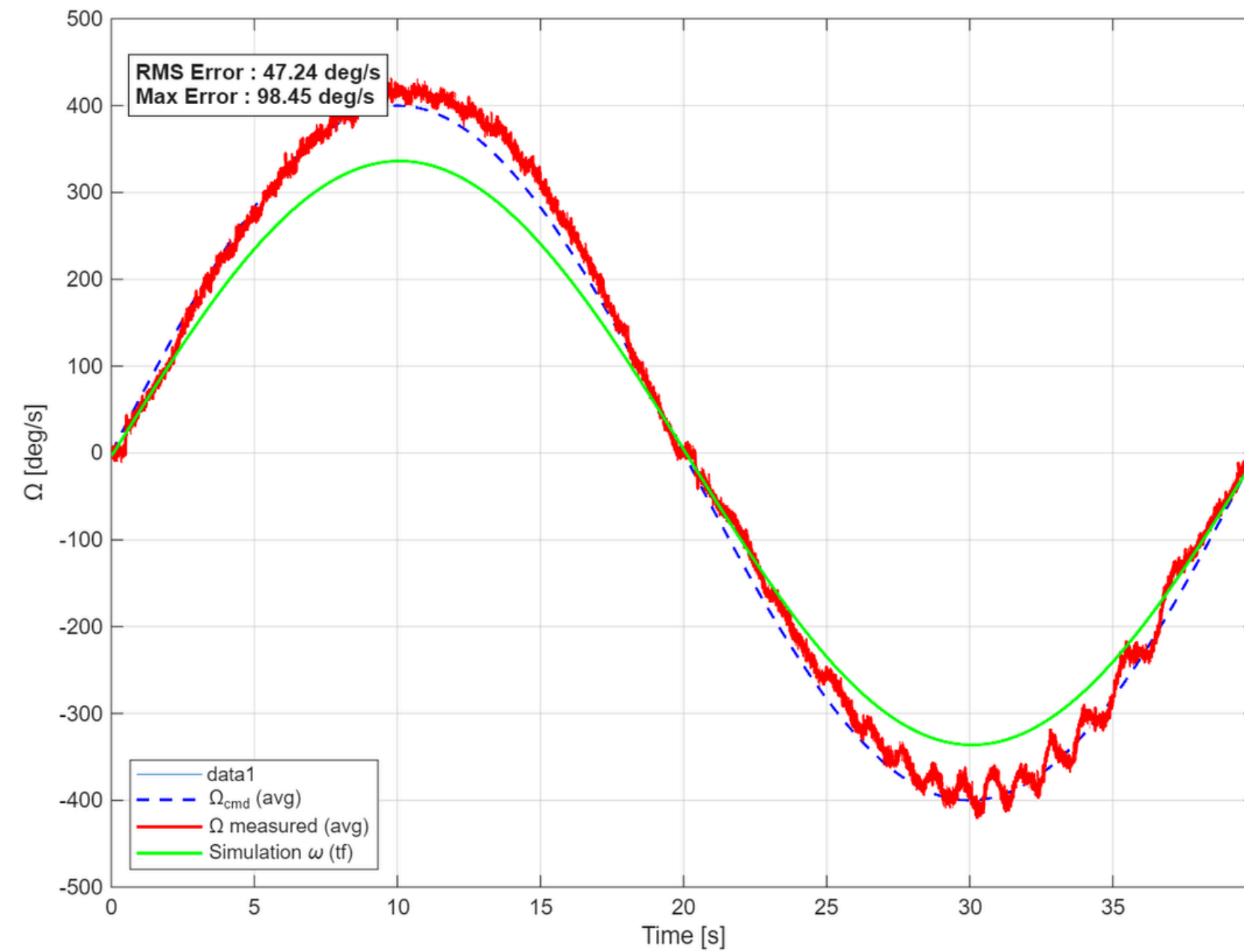
1차

deadzone omega 대략 38[deg]

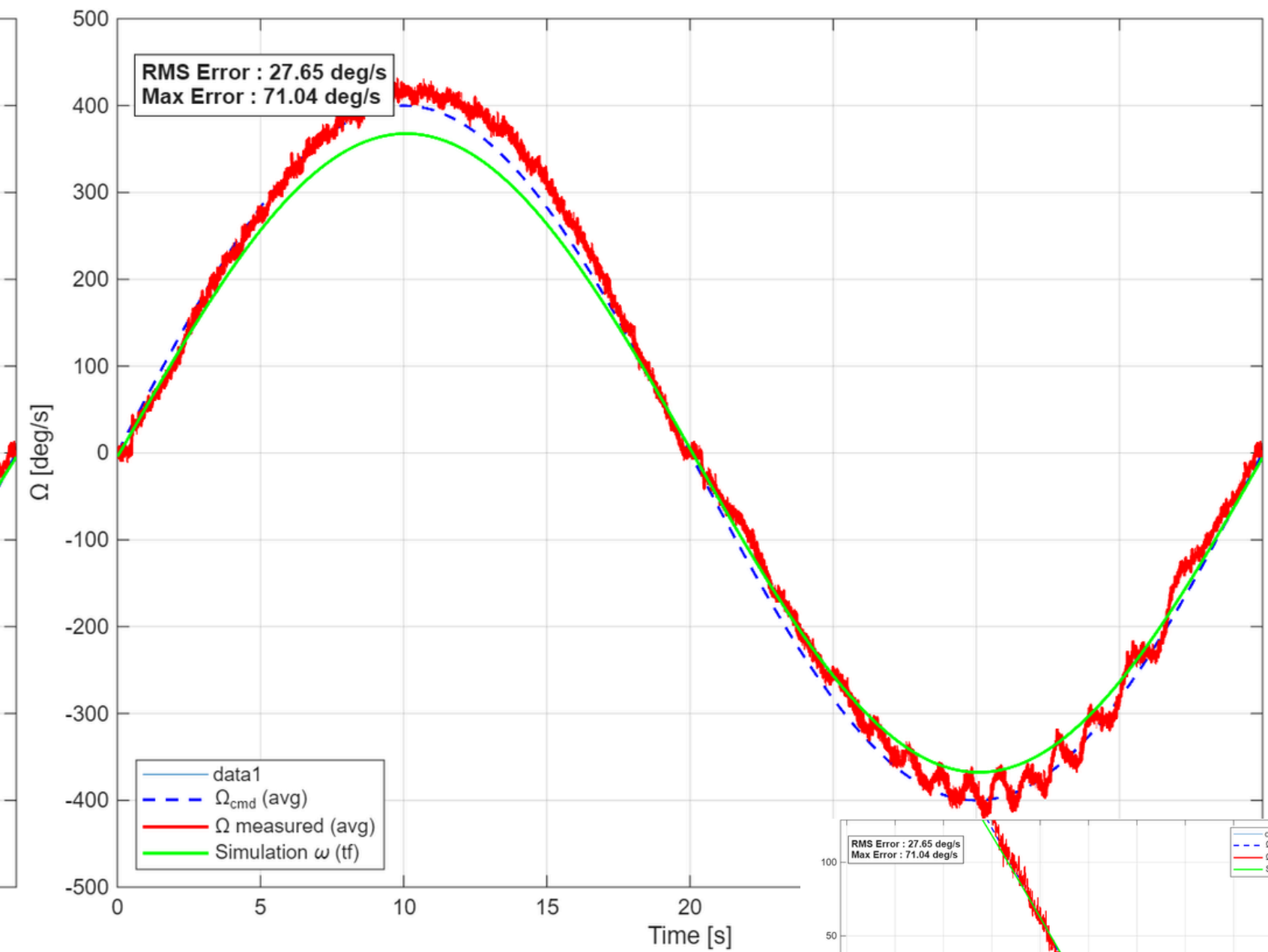


3. Model Validation

- 정현파 validation

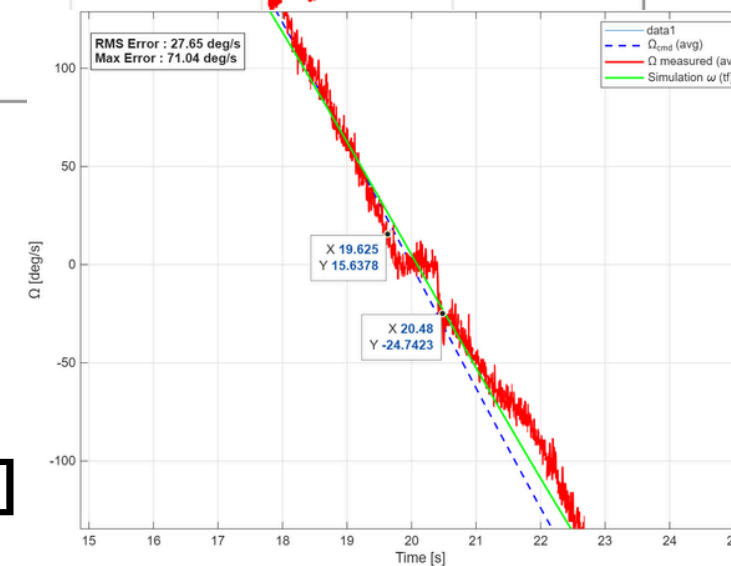


2차



1차

deadzone omega 대략 38[deg]



감사합니다